miniRT

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1 Camera

Given informations :

- $\mathbf{C_{pos}}$: 3D Position as a point.
- $\mathbf{C_{dir}}$: 3D Direction as a normalized vector.
- fov° : Vertical field of view in degrees, ranging from 0° to 180°.

Deduced informations :

- $\mathbf{C_{fov}}$: Vertical field of view in radians, ranging from 0 to π radians We can define it with the following fomula :

$$\mathrm{C_{fov}} = \mathrm{fov}^\circ \cdot \frac{\pi}{180^\circ}$$

- $\mathbf{C_{flen}}$: Focal length that can be calculated as follows :

$$\mathrm{C_{flen}} = \tan\left(\frac{\mathbf{C_{fov}}}{2}\right)$$

2 Ray intersections

2.1 Plane

2.2 Sphere

2.3 Circle

2.4 Cylinder